

Problem 29.7: Neutrino Oscillations

Source: M. D. Schwartz, *Quantum Field Theory and the Standard Model*, Problem 29.7.

Neutrino oscillations.

1. Neutrinos are produced in the Sun predominantly through the reaction $p + p + e^- \rightarrow d + \nu_e$. What is the momentum of the neutrinos produced this way?
2. Consider a two-neutrino flavour system. The mass eigenstates evolve in time as

$$\begin{aligned} |\nu_1\rangle &= e^{-iE_1 t} (\cos \theta |\nu_e\rangle + \sin \theta |\nu_\mu\rangle), \\ |\nu_2\rangle &= e^{-iE_2 t} (-\sin \theta |\nu_e\rangle + \cos \theta |\nu_\mu\rangle), \end{aligned} \quad (1.1)$$

where θ is the mixing angle. Show that the probability of finding a solar neutrino as an electron neutrino after a time T is given by

$$P = 1 - \sin^2(2\theta) \sin^2\left(\frac{(E_2 - E_1)T}{2}\right). \quad (1.2)$$

3. Take the high-energy limit $E \gg m_\nu$ to show that the probability of finding a solar neutrino with energy E as an electron neutrino at a distance L is given by

$$P = 1 - \sin^2(2\theta) \sin^2\left(\frac{\Delta m^2 L}{4E}\right). \quad (1.3)$$

4. How far should you put your detector from a reactor producing ~ 4 MeV neutrinos assuming $\Delta m^2 = 7.5 \times 10^{-5} \text{ eV}^2$ to see the largest effect?

Solution. Neglecting the thermal kinetic energy of the particles, the four-momentum before reaction is given by $k^\mu = (2m_p + m_e = s, 0)$. By four-momentum conservation $p_{\nu_e} + p_d = k$, we have

$$p_d^2 = m_d^2 = (k - p_{\nu_e})^2 = s + m_{\nu_e}^2 - 2\sqrt{s}E_{\nu_e} \implies E_{\nu_e} = \frac{s + m_{\nu_e}^2 - m_d^2}{s\sqrt{s}} = \sqrt{m_d^2 + |\mathbf{p}_{\nu_e}|^2}. \quad (1.4)$$

Therefore, we find

$$|\mathbf{p}_{\nu_e}| \approx 1.44 \text{ GeV}. \quad (1.5)$$

The solar neutrino is produced as an electron neutrino. That is, at $t = 0$,

$$|\nu; t = 0\rangle = |\nu_e\rangle = \cos \theta |\nu_1\rangle - \sin \theta |\nu_2\rangle. \quad (1.6)$$

Carrying out a standard QM calculation, it is easy to obtain (1.2).

In the high-energy limit,

$$E_1 \approx |\mathbf{p}| \left(1 - \frac{m_1^2}{2|\mathbf{p}|^2} \right) \implies -\Delta E \approx \frac{2\Delta m^2}{2|\mathbf{p}|} \approx \frac{2\Delta m^2}{2E}. \quad (1.7)$$

In natural units, $T \approx L$, and hence, we find the desired equation.

The maxima occur at

$$L_n = \frac{2\pi E}{\Delta m^2} (2n + 1), \quad L_0 \approx 66 \text{ km}. \quad (1.8)$$